

FLOWGUARD

Preventing Demurrage Losses at KPC Terminals

Board Investment Pitch

Deborah Omae • Data Engineer, FlowGuard Project Team

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Every Late Truck Puts Money at Risk

Gate queues, loading-bay shortages, and equipment slowdowns compound quietly — and are usually only noticed once a truck is already running late.

5

terminals exposed to
turnaround risk

87 min

average truck
turnaround today

KES 14.82M

demurrage exposure
seen in testing

Today, these problems are fixed only after they cause a delay — not before.

FlowGuard Sees the Delay Before It Happens

A decision-support platform, already built and tested, that watches terminal activity in real time and acts on it.

1

PREDICT

Forecasts each truck's expected gate-out time from live queue, bay, and equipment data.

2

DIAGNOSE & OPTIMIZE

Identifies the bottleneck and calculates the best fix using constrained optimization.

3

ACT, WITH GUARDRAILS

Low-risk fixes apply automatically; higher-impact ones wait for a human's one-click approval.

4

VERIFY & LOG

Every recommendation and outcome is recorded for audit and continuous improvement.

Already Validated — Not Just a Concept

FlowGuard is built end-to-end and was tested against 50 realistic terminal scenarios before this pitch.

92%

of at-risk situations
resolved (46 of 50)

87 → 56 min

average turnaround time,
a 36% improvement

KES 14.82M

in demurrage exposure
protected in testing

Next step: confirm these numbers with real terminal data in a live pilot.

The Financial Case for a Pilot

3-month live pilot, one terminal (Nairobi) — figures shown on a 12-month basis

Item	Amount (KES)
Estimated annual demurrage savings	14,820,000
One-time setup cost	3,500,000
Annual running cost	900,000
Net Year-1 benefit	10,420,000
Payback period	~3 months
Year-1 ROI	297%
3-Year ROI	617%

Why it pays back fast

Setup cost is recovered from avoided demurrage charges alone — the running cost is a small fraction of the annual savings, so almost all of the benefit drops to the bottom line from month one.

Two Risks We're Managing Deliberately

Live data may behave differently than test data

MITIGATION

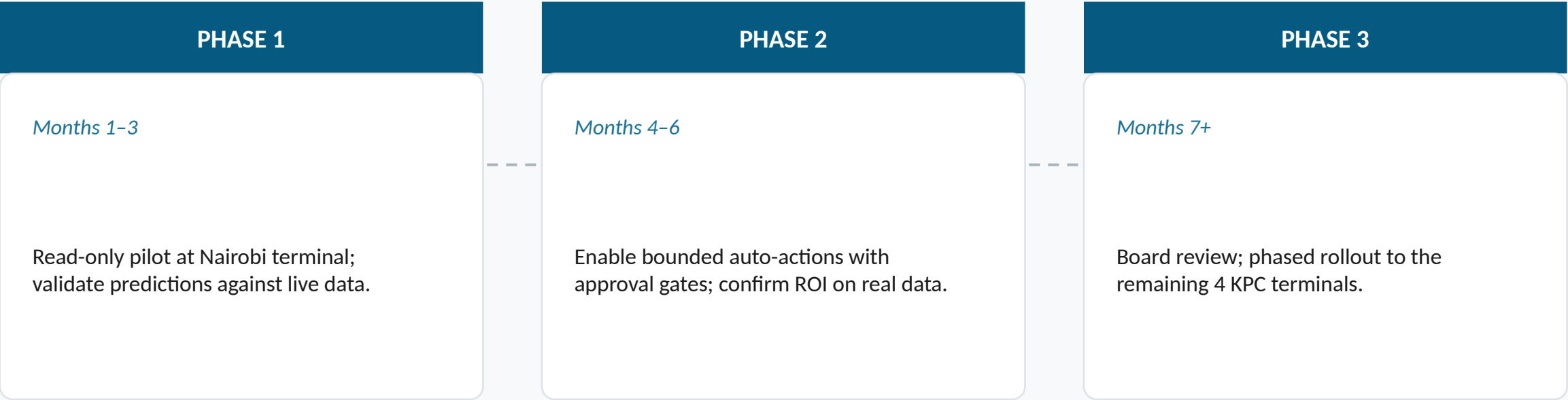
We start with a read-only 'watch and recommend' phase at one terminal before any automatic action is switched on, and recalibrate the model on real data before expanding.

Staff may be hesitant to trust automated recommendations

MITIGATION

Every higher-impact action requires a human's one-click approval. Recommendations are logged and explainable, and staff are trained before go-live.

A Phased, Low-Risk Path to Scale



The Ask

Approve a 3-Month Live Pilot

GO — KES 3.5M setup + KES 225K/quarter running cost. Expected payback in ~3 months if pilot results track pre-launch testing.

What we're asking the board to approve:

- Budget for a single-terminal, read-only pilot at Nairobi
- A full board review of live-pilot results before any wider rollout
- An executive sponsor to unblock live SCADA/ERP data access

Thank you — questions welcome.